MILITARY SPECIFICATION SHEET

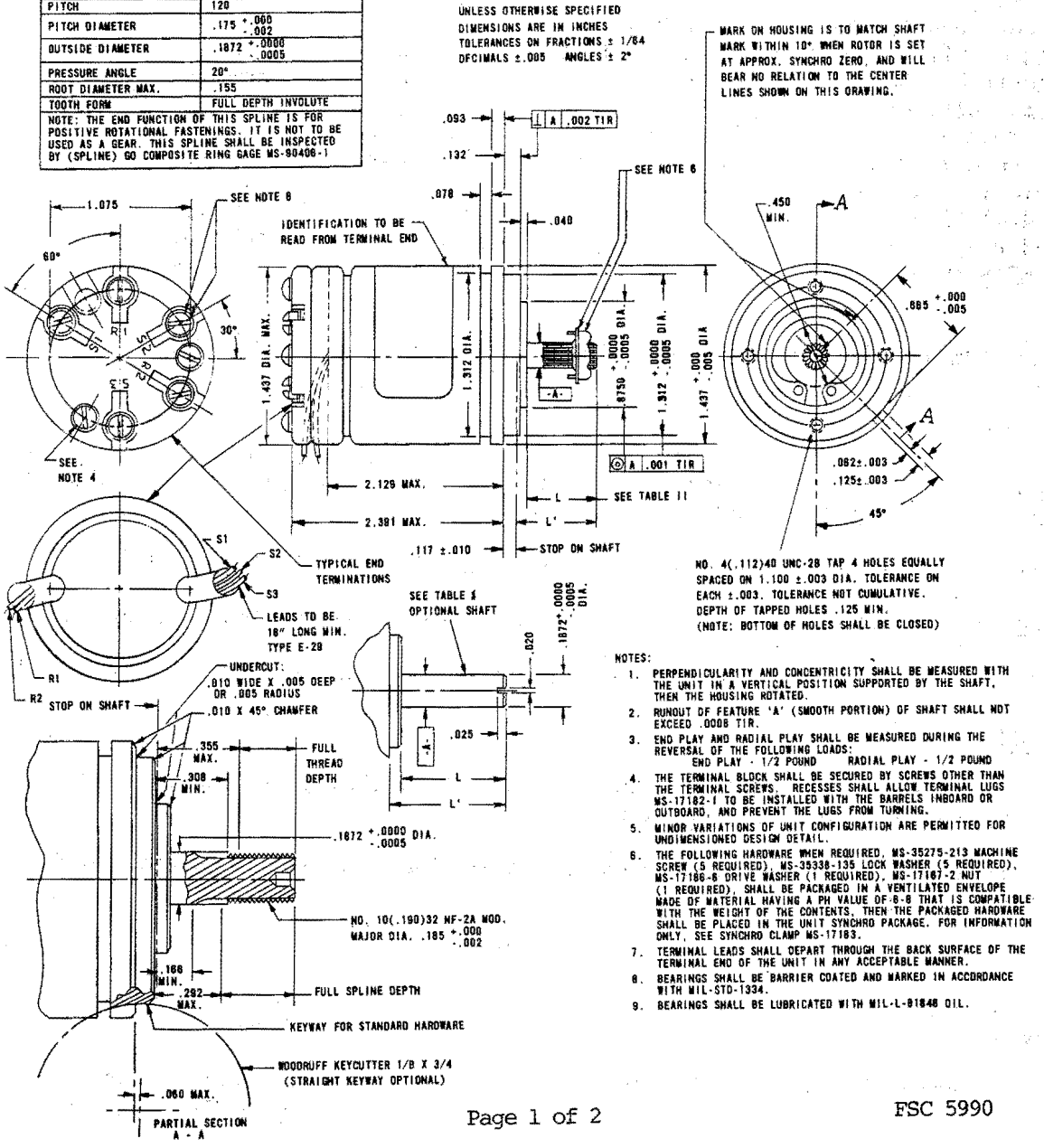
SYNCHROS, CONTROL TRANSFORMER, TYPE 15CT6B

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the control transformer synchros described herein shall consist of this document and the latest issue of specification MIL-S-20708.

SPLINE DATA	
TEETH	21
PITCH	120
PITCH DIAMETER	.175 +.000 -.002
OUTSIDE DIAMETER	.1872 +.0000 -.0005
PRESSURE ANGLE	20°
ROOT DIAMETER MAX.	.155
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE MS-90406-1



NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING ROTATED.
2. RUNOUT OF FEATURE 'A' (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0008 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:
   END PLAY - 1/2 POUND RADIAL PLAY - 1/2 POUND
4. THE TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS. RECESSES SHALL ALLOW TERMINAL LUGS MS-17182-1 TO BE INSTALLED WITH THE BARRELS INBOARD OR OUTBOARD, AND PREVENT THE LUGS FROM TURNING.
5. MINOR VARIATIONS OF UNIT CONFIGURATION ARE PERMITTED FOR UNDIMENSIONED DESIGN DETAIL.
6. THE FOLLOWING HARDWARE WHEN REQUIRED, MS-35275-213 MACHINE SCREW (5 REQUIRED), MS-35338-135 LOCK WASHER (5 REQUIRED), MS-17186-6 DRIVE WASHER (1 REQUIRED), MS-17187-2 NUT (1 REQUIRED), SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS. THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE. FOR INFORMATION ONLY, SEE SYNCHRO CLAMP MS-17183.
7. TERMINAL LEADS SHALL DEPART THROUGH THE BACK SURFACE OF THE TERMINAL END OF THE UNIT IN ANY ACCEPTABLE MANNER.
8. BEARINGS SHALL BE BARRIER COATED AND MARKED IN ACCORDANCE WITH MIL-STD-1334.
9. BEARINGS SHALL BE LUBRICATED WITH MIL-L-81846 OIL.

FSC 5990

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Requirement	Value	Unit	Tolerance
Frequency	60	Hz	±1%
Primary Voltage	90	volts	nominal
Primary Current	20-25	milliamps	min.-max.
Primary Power	0.7	watts	maximum
Impedance:			
Zso	3611	ohms	nominal
Zrs	1423	ohms	nominal
Impedance Angle:			
Zso	70.1	degrees	nominal
Zrs	15.9	degrees	nominal
Transformation Ratio	0.735	-------	±2%
Phase Shift	20.0 (lead)	degrees	±2.0
Electrical Error	6.0	minutes	maximum
Null Voltage:			
Total	90.0	millivolts	maximum
Fundamental	60.0	millivolts	maximum
Friction Torque	0.05	ounce-inches	maximum
Radial Play	0.0004	inches	maximum
End Play	0.0015	inches	maximum
Temperature Rise	10.0	degrees (C)	maximum

Part number: M20708/504 plus applicable dash number from Table I.

TABLE I

DASH NO.	SHAFT DIAMETER	SHAFT STYLE	TERMINAL END	L' DIMENSION TO STOP ON SHAFT	L REFERENCE FREE SHAFT LENGTH
-001	.1872	Splined	Terminal Board	.555 ± .010	.500 ± .010
002	.1872	Smooth	Terminal Board	.555 ± .010	.500 ± .010
003	.1872	Smooth	Leads	.555 ± .010	.500 ± .010

NOTES:

1. This component used in Surface Missile System (SMS), Missile Fire Control System (MFCS), MK 115.
2. This synchro is for replacement purposes only and is not recommended for new design.

Custodian:
Navy-OS

Preparing Activity
Navy-OS

Project Number
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